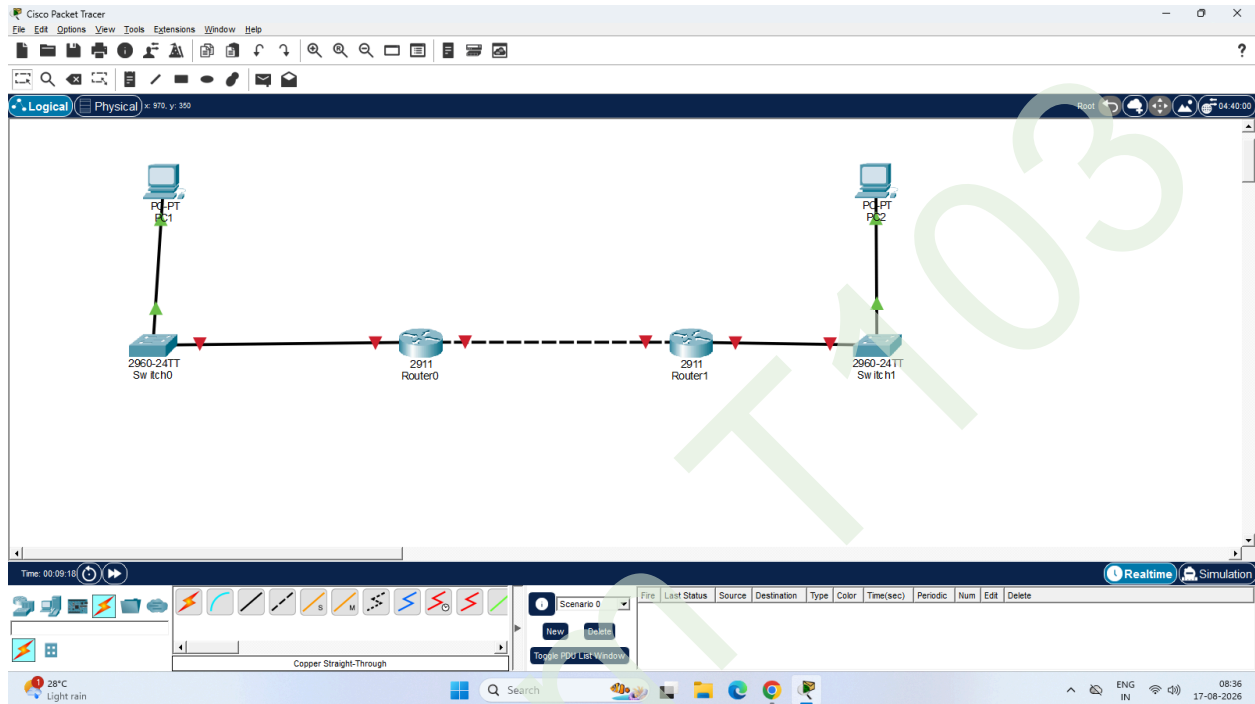


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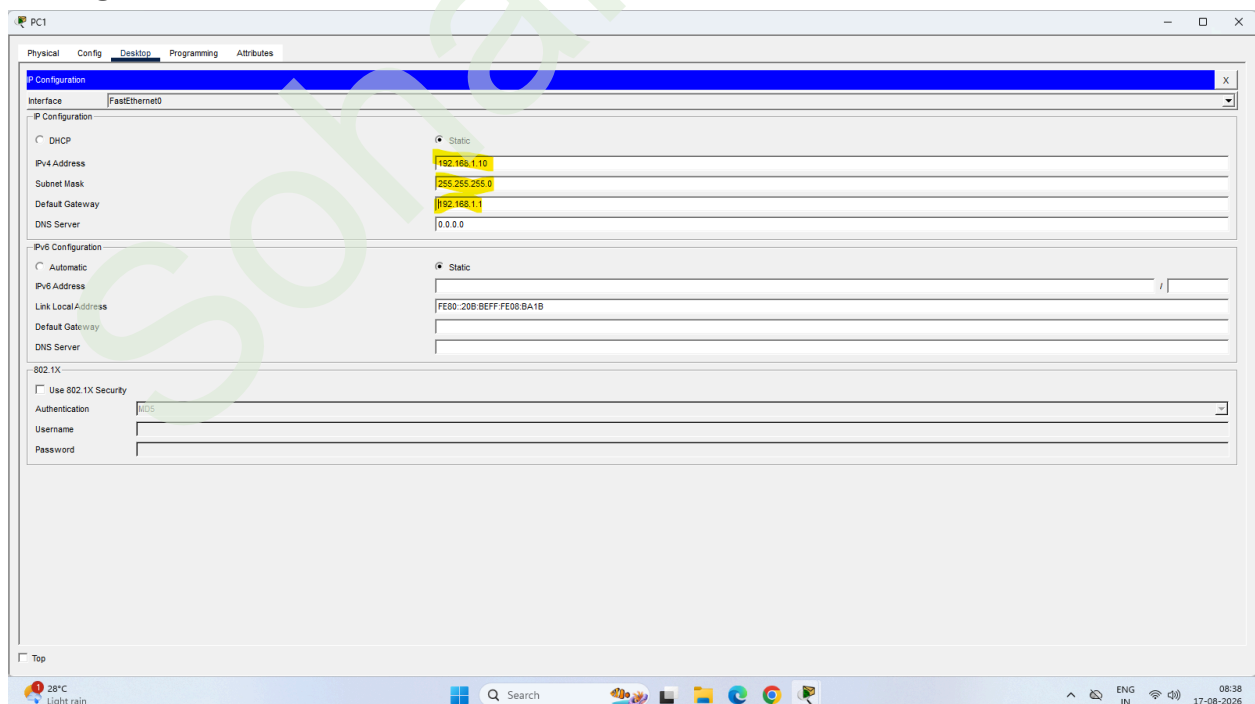
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Aim: Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

Create the Topology



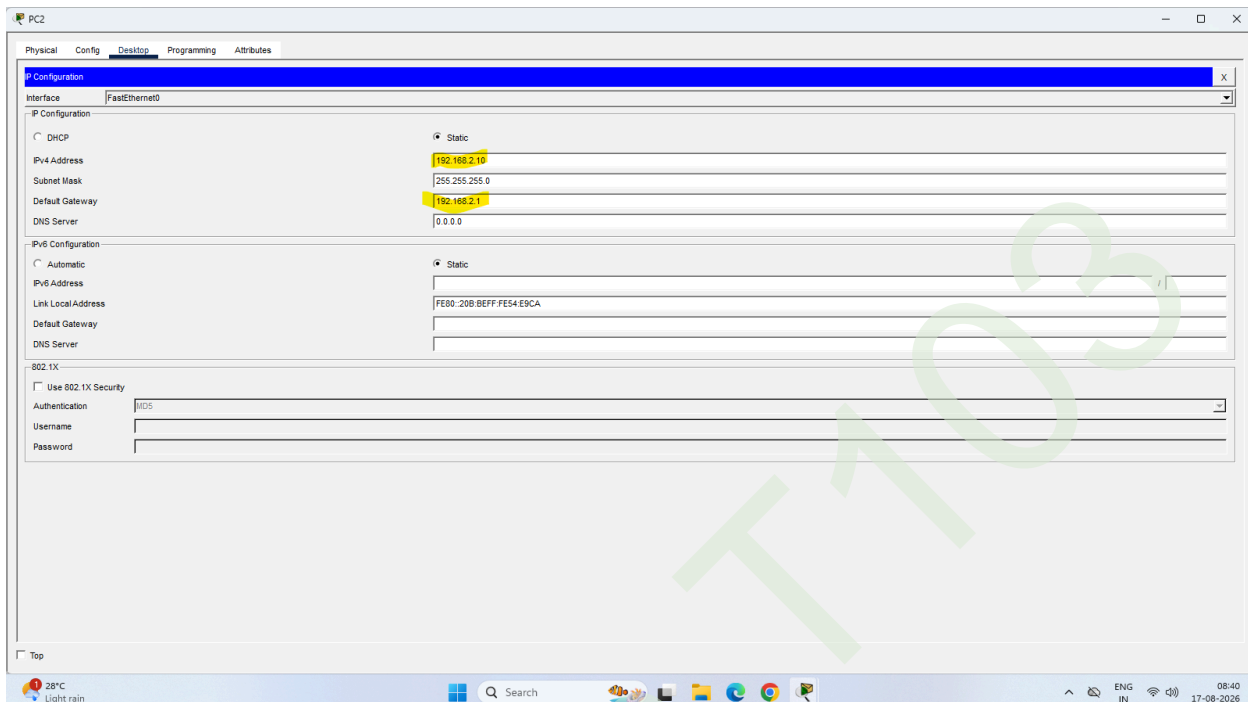
Configure PC1



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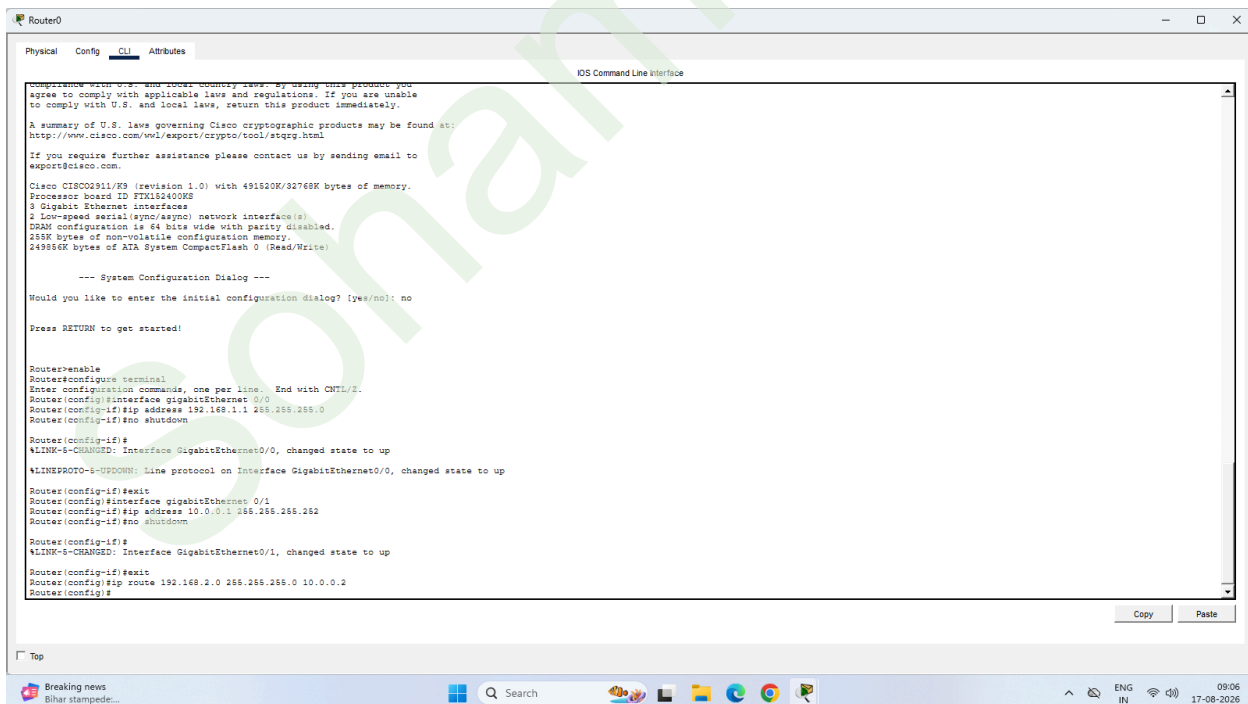
SUBJECT : Cyber & Information Security

Configure PC2



The screenshot shows the 'PC2' configuration window with the 'Config' tab selected. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The 'IPv4 Address' is set to '192.168.2.10', 'Subnet Mask' is '255.255.255.0', and 'Default Gateway' is '192.168.2.1'. The 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', the 'Static' radio button is selected, and the 'Link Local Address' is 'FE80:20B:BEFF:FE54:EBCA'. The '802.1X' section is collapsed. The Windows taskbar at the bottom shows the date as 17-08-2026 and time as 08:40.

Configure Router0



The screenshot shows the 'Router0' configuration window with the 'CLI' tab selected. The 'IOS Command Line Interface' is displayed. The text in the CLI window is as follows:

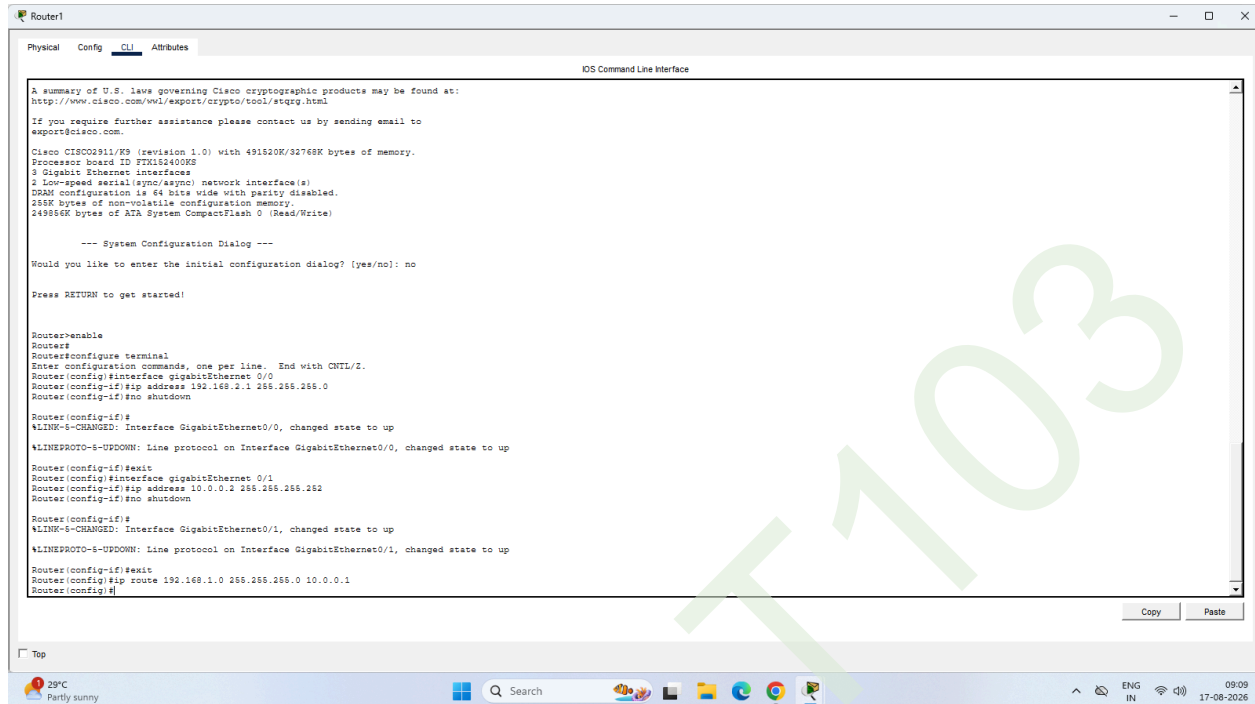
```
compliance with U.S. and other country laws. By using this product you agree to comply with applicable laws and regulations. If you are unable to comply with U.S. and local laws, return this product immediately. A summary of U.S. laws governing Cisco cryptographic products may be found at: http://www.cisco.com/wml/export/crypto/cool/stgcy.html If you require further assistance please contact us by sending email to export@cisco.com. Cisco C1800D01/02 (revision 1.0) with 491520K/32768K bytes of memory. Processor board ID FTK15240002 3 Gigabit Ethernet interfaces 2 Low-speed serial (syn/asyn) network interfaces DRAM configuration is 64 bits wide with parity disabled. 256K bytes of non-volatile configuration memory. 249956K bytes of ATA System CompactFlash 0 (Read/Write) --- System Configuration Dialog --- Would you like to enter the initial configuration dialog? [yes/no]: no Press RETURN to get started! Router>enable Router>configure terminal Enter configuration commands, one per line. End with CNTL/Z. Router(config)#interface gigabitEthernet 0/0 Router(config-if)#ip address 192.168.1.1 255.255.255.0 Router(config-if)#no shutdown Router(config-if)# %LINK-3-CHANGED: Interface GigabitEthernet0/0, changed state to up %LINEPROTO-3-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up Router(config-if)#exit Router(config)#interface gigabitEthernet 0/1 Router(config-if)#ip address 10.0.0.1 255.255.255.252 Router(config-if)#no shutdown Router(config-if)# %LINK-3-CHANGED: Interface GigabitEthernet0/1, changed state to up Router(config-if)#exit Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2 Router(config)#
```

The Windows taskbar at the bottom shows the date as 17-08-2026 and time as 08:06.

Configure Router1

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```
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wol/exports/crypto/ascii/rtspz.htm

If you require further assistance please contact us by sending email to
export@cisco.com.

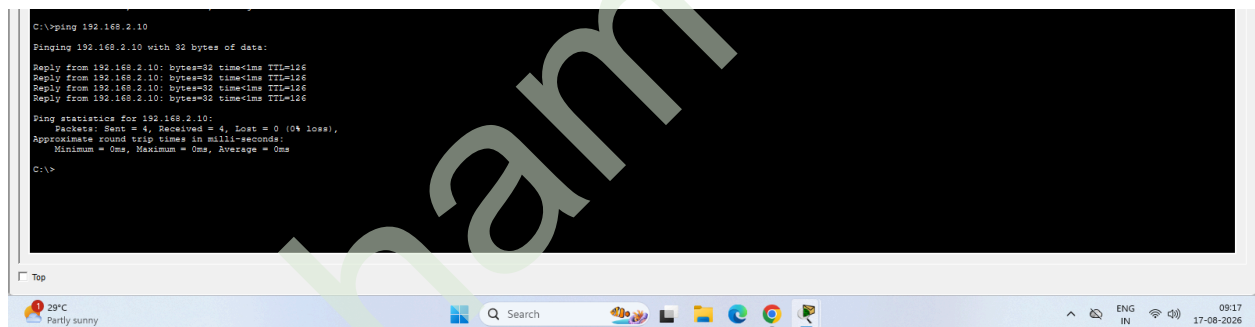
Cisco IOS02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTK152400K2
3 Gigabit Ethernet interfaces
2 low-speed serial (sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249584K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#
```



```
C:\>ping 192.168.2.10

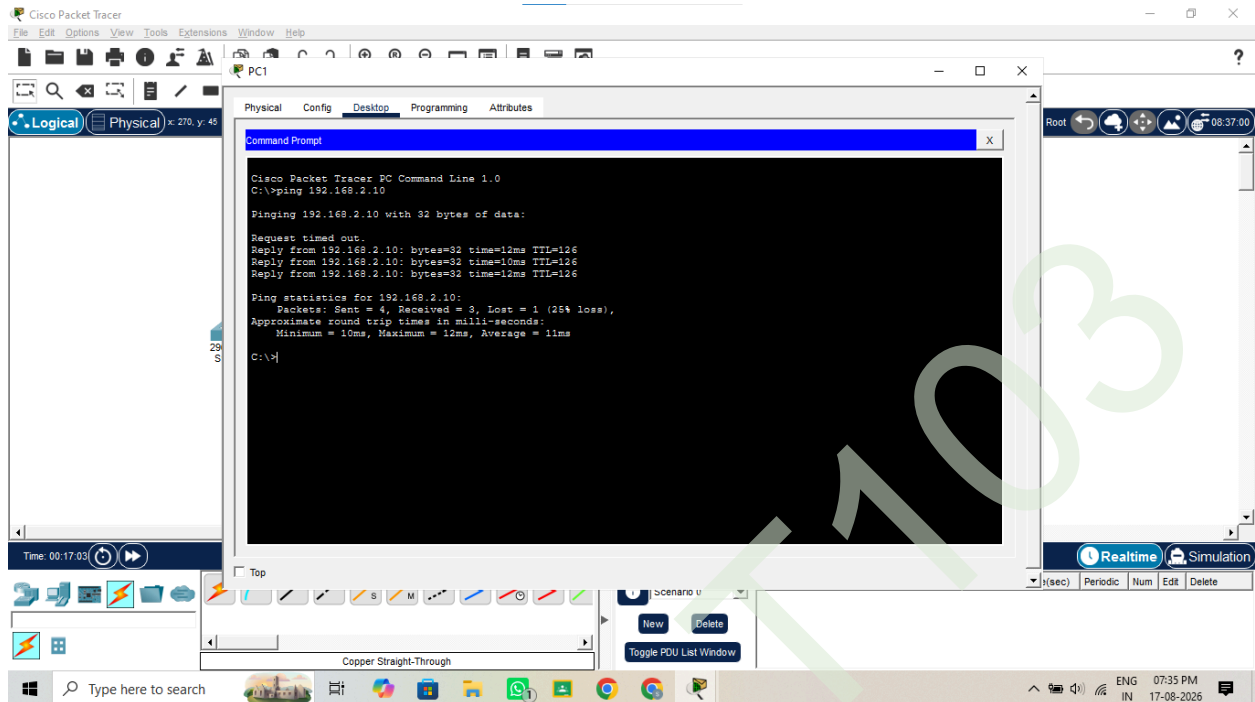
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time=126ms TTL=126
Reply from 192.168.2.10: bytes=32 time=126ms TTL=126
Reply from 192.168.2.10: bytes=32 time=126ms TTL=126
Reply from 192.168.2.10: bytes=32 time=126ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

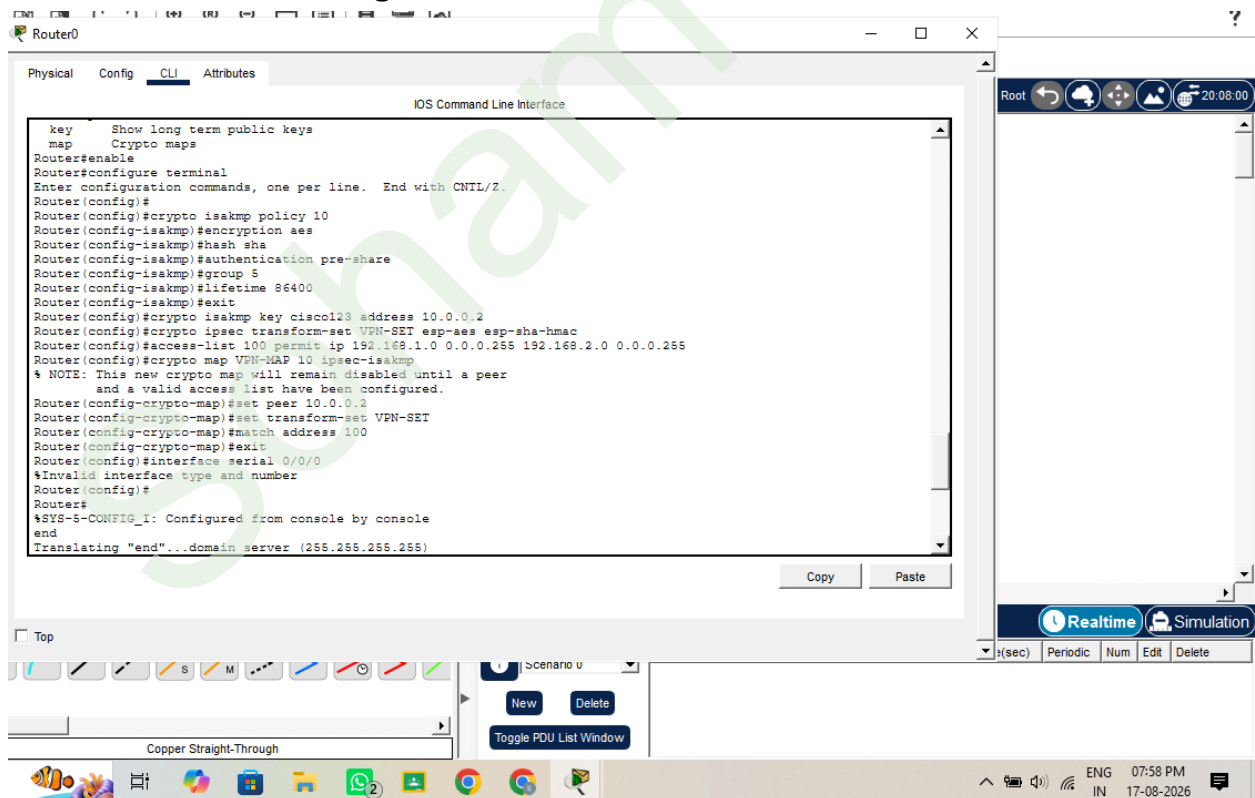
C:\>
```

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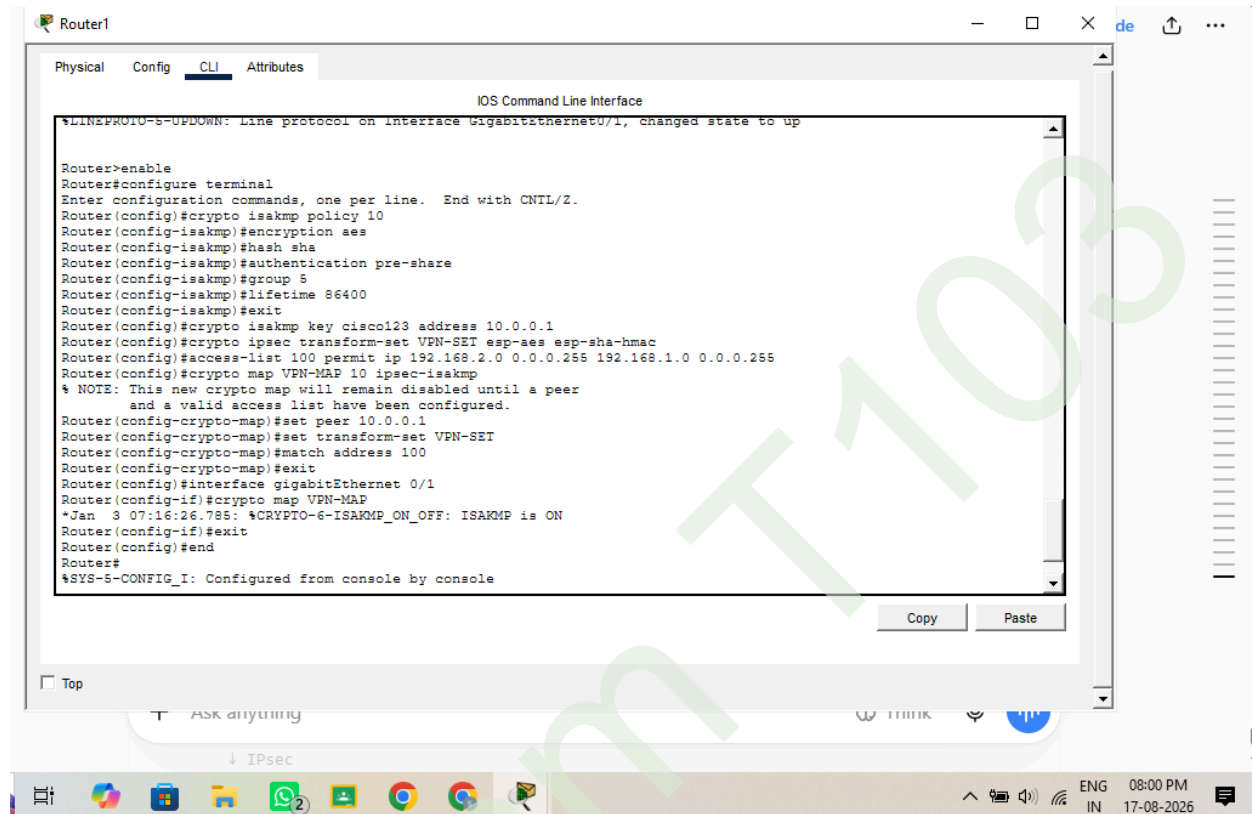
ROUTER 0 IPsec configurations



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Router 1 IPsec configurations



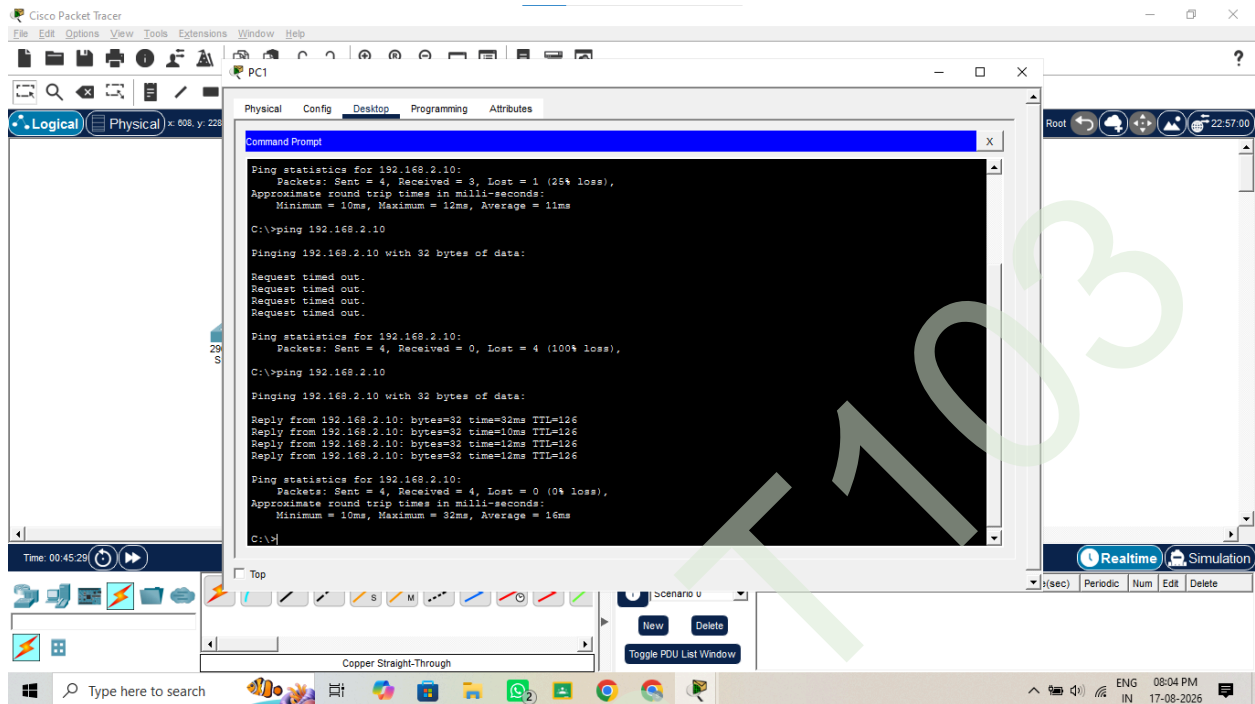
```
Router1
Physical Config CLI Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on interface GigabitEthernet0/1, changed state to up

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

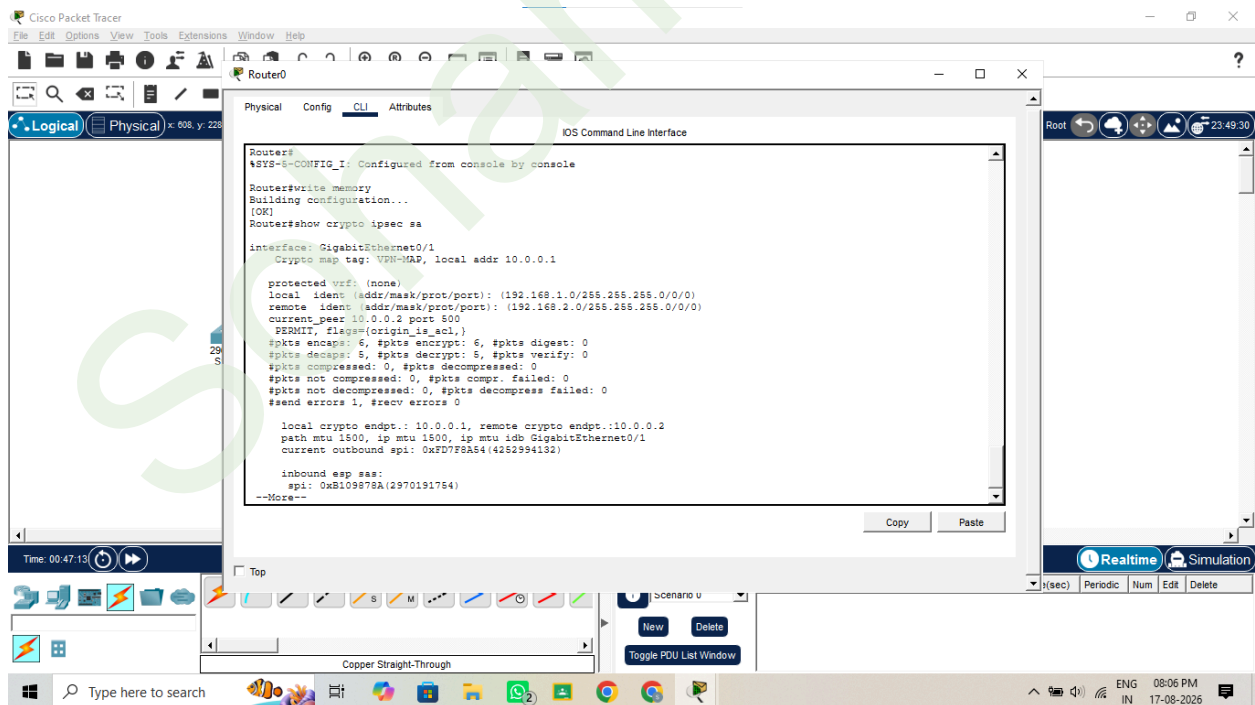
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PC1 → Desktop → Command Prompt



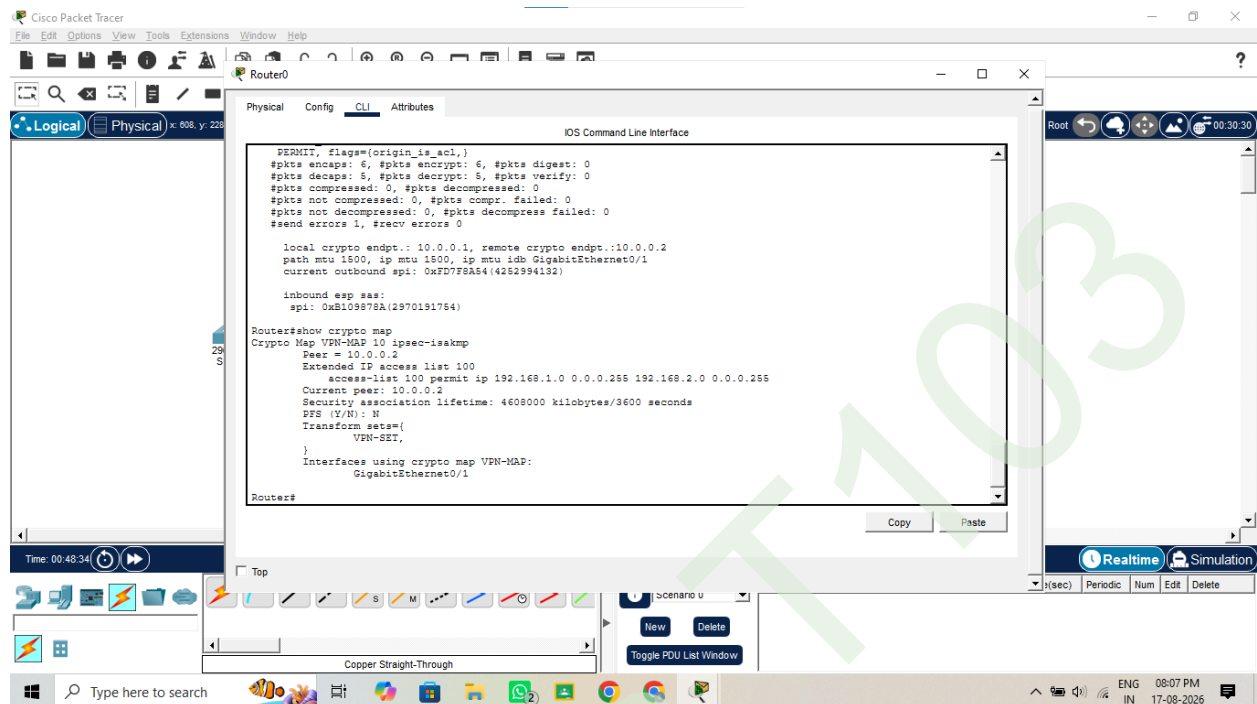
Check IPsec



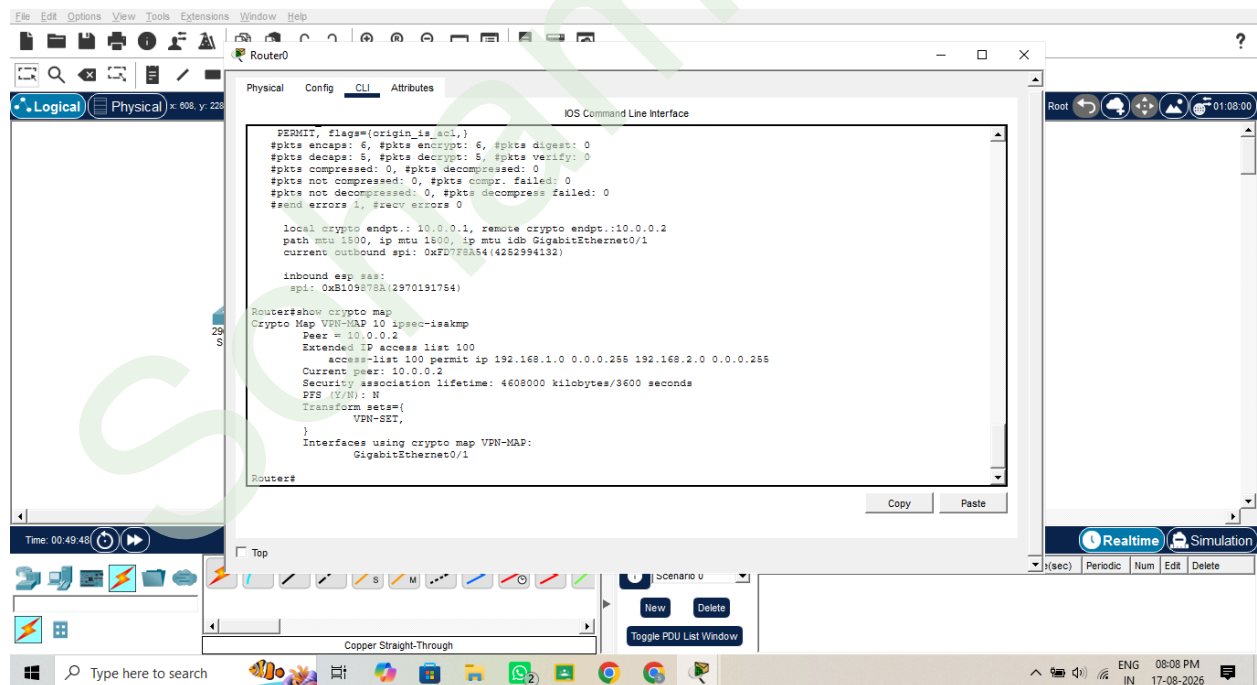
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Check Crypto Map



Check Router1

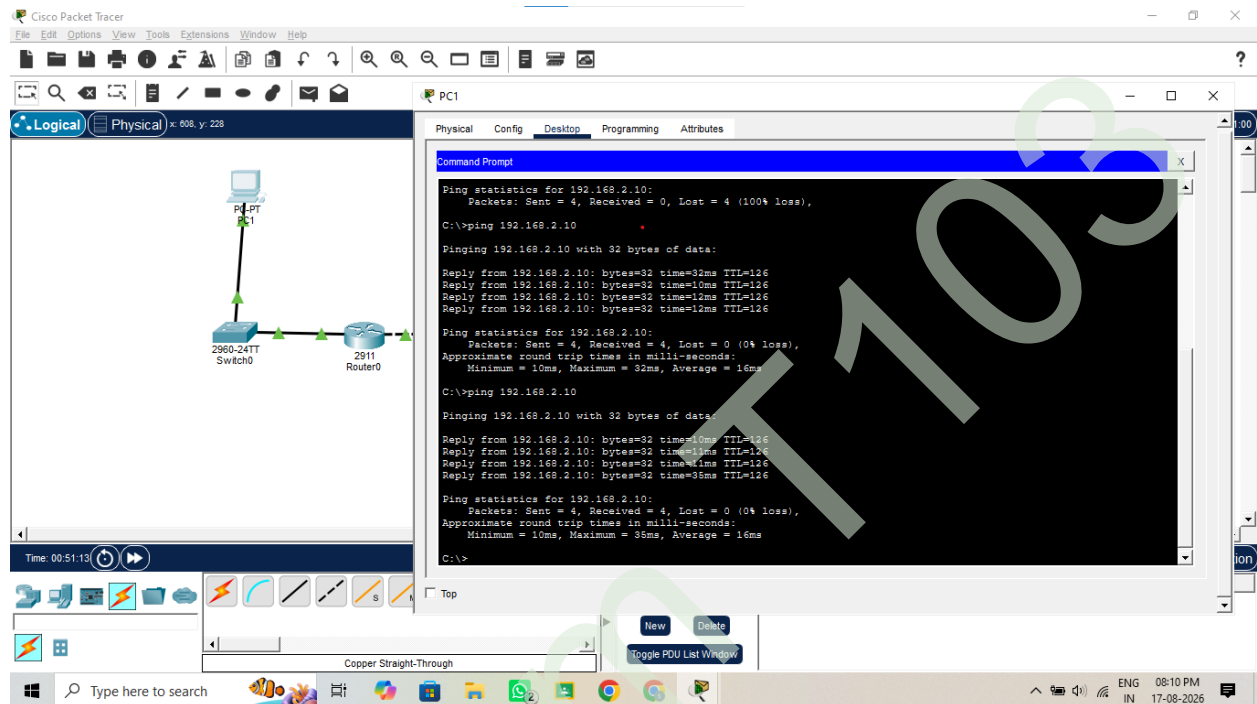


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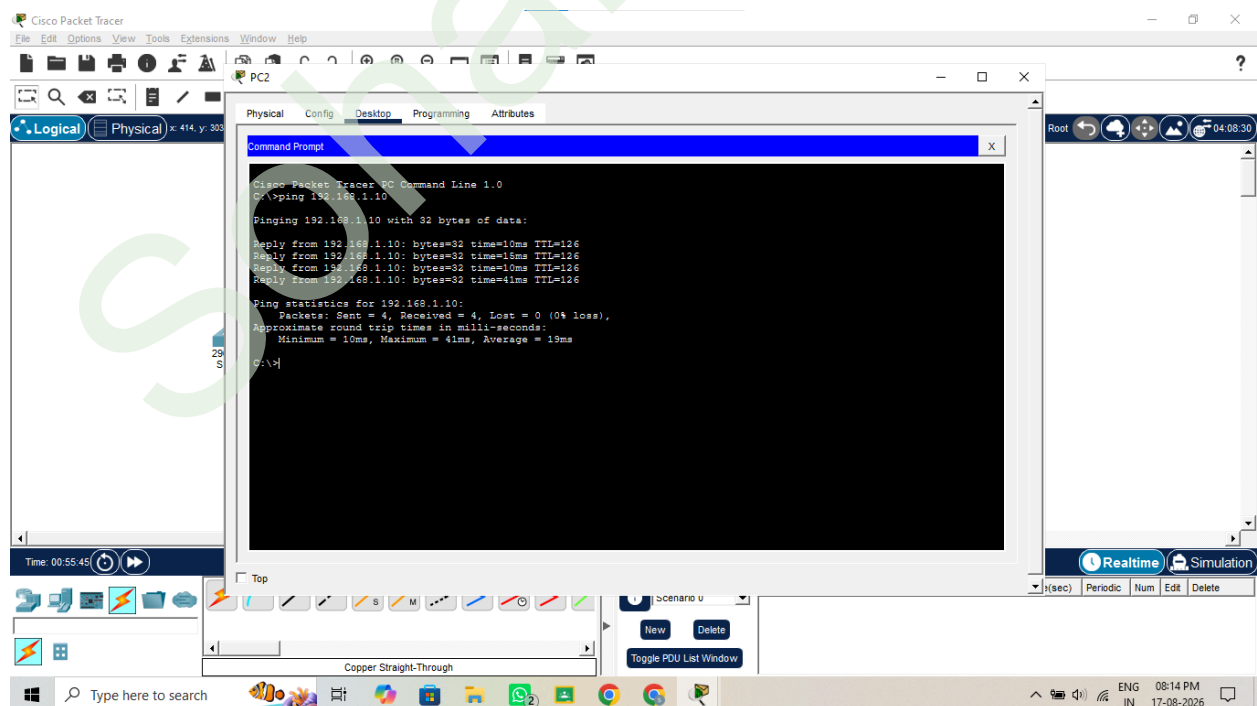
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Final Test

Finally, from PC1



PC2



Soham Patil
T103 TyCs